

Biomechanical Comparison of Implants for Intertrochanteric Fracture Fixation Using Stress Analysis

*Project report submitted to
Visvesvaraya National Institute of Technology, Nagpur
in partial fulfillment of the requirements for the award of
the degree*

Bachelor of Technology in Mechanical Engineering

by

Aditi Verma (BT22MEC002) Sanskriti Rathore (BT22MEC010)

under the guidance of
Dr. Rashmi Uddanwadiker



Department of Mechanical Engineering
Visvesvaraya National Institute of Technology
Nagpur 440010 (India)

April 2026

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April 2026



Declaration

We, **Sanskriti Rathore (BT22MEC010)** and **Aditi Verma (BT22MEC002)**, hereby declare that this project work titled **“Biomechanical Comparison of Implants for Intertrochanteric Fracture Fixation Using Stress Analysis”** is carried out by us in the **Department of Mechanical Engineering** of Visvesvaraya National Institute of Technology, Nagpur. The work is original and has not been submitted earlier whole or in part for the award of any degree/diploma at this or any other Institution / University.

Sr.No.	Enrollment No	Names	Signature
1.	BT22MEC010	Sanskriti Rathore	
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Date:

Certificate

This is to certify that the project titled **“Biomechanical Comparison of Implants for Intertrochanteric Fracture Fixation Using Stress Analysis”**, submitted by **Sanskriti Rathore** and **Aditi Verma** in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in **Mechanical Engineering**, VNIT Nagpur. The work is comprehensive, complete and fit for final evaluation.

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Acknowledgement

We would like to convey our deepest sense of appreciation to all the individuals and organizations that have guided and supported us during our project, entitled “**Biomechanical Comparison of Implants for Intertrochanteric Fracture Fixation Using Stress Analysis.**” The entire project was a wonderful learning experience for us, and we owe our success to many individuals and organizations.

We first acknowledge the contributions of our project guide, **Dr. Rashmi Uddanwadiker**, for her invaluable guidance, encouragement, and insight. We are extremely grateful to her for her immense expertise in mechanical engineering and biomedical engineering, as well as for the excellent suggestions she provided for our project. Her support not only enabled us to improve our approach but also inspired us to be innovative and analytical throughout the project.

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Finally, we express our sincere gratitude to our parents, family members, and friends for their constant encouragement, motivation, and moral support throughout the project duration.

Abstract

This study presents a biomechanical comparison of common implants used for the fixation of intertrochanteric femur fractures using Finite Element Analysis (FEA). Intertrochanteric fractures, mostly seen among elderly patients due to osteoporosis, have challenges in terms of stability, load distribution, and risk of implant failure. The objective of this work is to simulate and compare the mechanical performance of extramedullary and intramedullary implants, including Dynamic Hip Screw (DHS), Angled Blade Plate, Proximal Femoral Nail (PFN), PFNA-II, and InterTAN.

A 3D CAD model of a healthy femur of a 60 year old male with 75 kgs of weight was developed to simulate an intertrochanteric fracture based on clinical inputs and X-ray references. Each implant was modeled and assembled with the fractured femur. Material properties for cortical bone, cancellous bone, and titanium alloy were assigned based on literature. Physiological loading conditions corresponding to weight on a one-legged stance during fractured and healing was taken into consideration.

We performed Finite Element Analysis to determine stress distribution across different implant assemblies under fractured and healed conditions. We found that the results indicate variations in stress concentration among implants, differentiating the implant designs on load sharing and stability. Intramedullary implants showcased similar load distribution to healthy bone without fracture, whereas certain assemblies showed a higher stress average and did not have similar stress distribution compared to healthy bone without fracture.

This study compares the biomechanical performance of intertrochanteric fixation implants and helps in finding the optimal implant for enhanced stability and reduced risk of failure in intertrochanteric fracture treatment.

Keywords: Intertrochanteric fracture, Femur, Finite Element Analysis, DHS, PFN, PFNA-II, InterTAN, Angled Blade Plate, Orthopedic Implants, Stress Distribution, Biomechanics.

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List of Publications

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Abstract - This research presents a comparative biomechanical investigation of commonly used orthopedic implants for fixation of intertrochanteric femur fractures using stress analysis and Finite Element Analysis (FEA). The implants considered in the study include Dynamic Hip Screw (DHS), Angled Blade Plate, Proximal Femoral Nail (PFN), PFNA-II, and InterTAN. A three-dimensional femur model with simulated fracture geometry was developed and each implant was assembled under clinically relevant conditions. Material properties corresponding to cortical bone, cancellous bone, and titanium alloy were assigned. Physiological loading representing one-legged stance was applied to evaluate stress distribution, deformation, and fixation stability. The study aims to identify the most efficient implant design with improved load sharing, reduced stress concentration, and enhanced mechanical stability for treatment of intertrochanteric fractures.

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Dear Ms. Sanskriti Rathore,

Warm Greetings!!

Congratulations! Your research article titled "Biomechanical Comparison of Implants for Intertrochanteric Fracture Fixation Using Stress Analysis" has been accepted for presentation at International Conference on Bioinformatics Applications in Biomedical Innovations (ICBABI-26) scheduled for 30th May 2026 | Visakhapatnam - India.

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CHAPTER 1

INTRODUCTION

The femur is the longest, strongest, and one of the most important load-bearing bones in the human body [1]. It forms the skeletal connection between the hip joint and knee joint and is responsible for transmitting body weight during standing, walking, running, stair climbing, and other daily activities [2]. Due to its structural role, the femur is continuously subjected to compression, bending, torsion, and shear forces [3].

Anatomically, the femur is divided into three main regions: proximal femur, shaft, and distal femur. The proximal femur consists of the femoral head, femoral neck, greater trochanter, and lesser trochanter [1]. The femoral head articulates with the acetabulum of the pelvis to form the hip joint, while the neck connects the head to the shaft as shown in 1.1. The greater and lesser trochanters act as major attachment sites for muscles controlling hip motion and stability [4].

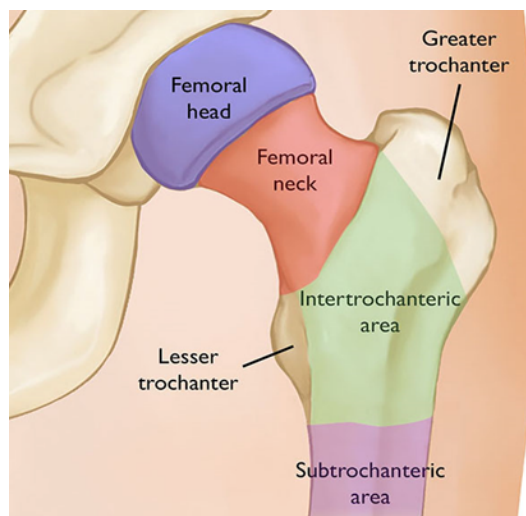


Figure 1.1: Anatomy of proximal femur showing greater and lesser trochanter

1.1 Types of Femoral Fractures

Femoral fractures may occur due to trauma, falls, osteoporosis, sports injuries, or road accidents [5]. Depending on the anatomical location, femoral fractures are generally classified into the following types:

- **Femoral Head Fractures:** These fractures involve the spherical head of the femur which articulates with the hip socket. They are usually associated with high-energy trauma such as hip dislocations or road traffic accidents [4]. Because they involve the articular surface, they are clinically serious and may lead to avascular necrosis or arthritis.

- **Femoral Neck Fractures:** These occur in the narrow region between the femoral head and the trochanteric area. They are common in elderly osteoporotic patients after minor falls [6]. Blood supply to the femoral head may be disrupted, making healing difficult and increasing risk of non-union.
- **Intertrochanteric Fractures:** These fractures occur between the greater trochanter and lesser trochanter of the femur. They account for a significant proportion of all proximal femoral fractures and are common in elderly individuals who usually require internal fixation [6]. Due to muscular forces and poor bone quality, maintaining stability can be challenging [7].
- **Subtrochanteric Fractures:** These occur just below the lesser trochanter, generally within 5 cm of the proximal shaft. This region is subjected to high compressive and tensile stresses, making fixation difficult [4]. These fractures are often unstable and may require intramedullary nailing.
- **Shaft Fractures:** These fractures occur along the long cylindrical middle portion of the femur. They usually result from high-energy trauma such as road accidents [1]. Significant blood loss, pain, and soft tissue damage may accompany such injuries.
- **Distal Femur Fractures:** These fractures occur near the lower end of the femur close to the knee joint. They are common in elderly osteoporotic patients and in high-energy trauma cases [1]. If the fracture extends into the knee joint, restoration of alignment becomes essential to preserve knee function.

Among these, fractures occurring in the proximal femur, especially femoral neck and intertrochanteric fractures, are clinically significant because they directly affect mobility, weight transfer, and hip joint function [6].

1.2 Intertrochanteric Fracture

Intertrochanteric fractures occur in the region between the greater trochanter and lesser trochanter of the femur as shown in 1.2. These fractures are commonly seen in elderly individuals suffering from osteoporosis, where bone density decreases significantly [6]. In such patients, even a simple fall from standing height may result in fracture [7]. In younger adults, they are usually caused by high-energy trauma [4]. The annual number of intertrochanteric fractures is projected to rise substantially with the ageing global population, with worldwide hip fracture incidence estimated to exceed 6 million by 2050 [8].

Intertrochanteric fractures are highly challenging because the proximal femur experiences large compressive and shear forces during gait. During the single-legged stance phase of gait, the hip joint reaction force is approximately 2.5–3 times body weight, directed obliquely through

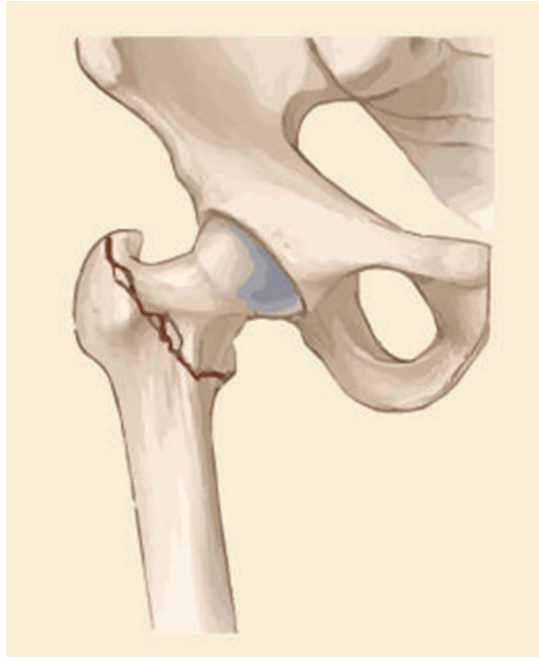


Figure 1.2: Intertrochanteric fracture region of femur

the femoral neck [9]. Once fractured, the natural load path is disturbed and the fragments tend to displace under body weight and muscular pull [4].

The **lesser trochanter** is of particular clinical importance because it serves as the insertion point of the iliopsoas muscle, one of the strongest hip flexor muscles [6]. When the lesser trochanter fragment is broken, the pull of the iliopsoas displaces the fragment medially and proximally, making reduction difficult [5]. In many unstable fractures, loss of lesser trochanter support also weakens the medial cortical buttress, reducing structural stability and increasing chances of implant failure, collapse, or malunion [10].

The one-year mortality rate following intertrochanteric fracture has been reported at approximately 17% on average, with rates varying by patient age and comorbidity burden [8]. Therefore, restoration of alignment and stable fixation is essential for successful healing and survival of these patients.

1.3 Need for Internal Fixation Implants

Conservative treatment of intertrochanteric fractures often leads to prolonged bed rest, poor union, muscle wasting, and life-threatening complications such as deep vein thrombosis and pulmonary embolism in elderly patients [6]. Hence, surgical fixation using orthopedic implants is the standard of care.

These implants are designed to [4, 7]:

- Stabilize fractured bone fragments
- Restore anatomical alignment

- Transfer physiological loads safely
- Allow early mobilization
- Promote fracture healing
- Reduce risk of secondary displacement

Over time, several implant systems were developed to overcome the limitations of earlier designs, evolving from rigid plates to dynamic and intramedullary constructs [7].

1.4 Types of Fixation Implants

Based on placement and load transfer mechanism, implants used for intertrochanteric fracture fixation are broadly classified into two categories: **extramedullary implants** and **intramedullary implants** [4, 6].

1.4.1 Extramedullary Implants

Extramedullary implants are fixed on the outer surface of the femur, generally along the lateral cortex. These devices stabilize the fracture externally using plates, screws, or blades that connect the proximal fragment to the femoral shaft [4]. Since they are positioned away from the natural mechanical axis of the femur, they are subjected to comparatively higher bending moments during loading [7]. However, they are relatively simple to implant, economical, and remain effective for stable fracture patterns [6].

Due to their lateral placement, extramedullary devices rely heavily on the integrity of the lateral femoral wall and bone quality for maintaining stability. In cases of unstable fractures or osteoporotic bone, the increased bending moment can lead to complications such as implant failure, excessive collapse, or varus deformity. Despite these limitations, they continue to be widely used in clinical practice due to their ease of surgical technique, lower cost, and well-established clinical outcomes.

Common examples of extramedullary fixation devices are the Dynamic Hip Screw and the Angled Blade Plate.

Dynamic Hip Screw (DHS)

The Dynamic Hip Screw was developed to overcome the limitations of rigid fixed-angle plates by allowing controlled sliding and compression at the fracture site [6]. It consists of a lag screw inserted into the femoral head connected to a lateral side plate fixed to the shaft. During weight bearing, the lag screw can slide within the barrel, producing fracture impaction and promoting healing [7].

DHS is widely used because of its reliable clinical outcomes, simpler surgical technique, and lower cost [4]. It is generally preferred in stable intertrochanteric fractures (AO/OTA type

31-A1) with good bone quality [8]. However, in unstable fractures or severe osteoporosis, complications such as screw cut-out, excessive collapse, or varus deformity may occur [7, 10].

The Dynamic Hip Screw implant is shown in Fig. 1.3.



Figure 1.3: Dynamic Hip Screw (DHS)

Angled Blade Plate

The Angled Blade Plate is a rigid fixation device developed to provide fixed-angle stability and accurate control of fracture alignment [7]. It consists of a blade inserted into the femoral head and neck, attached to a side plate fixed on the shaft.

This implant provides strong angled support and is useful in selected unstable fractures, reverse obliquity patterns, revision surgeries, and corrective procedures [6]. It offers excellent alignment control but requires precise surgical technique during insertion. Due to technical difficulty and the availability of modern intramedullary nails, it is less commonly used today [4].

The Angled Blade Plate is shown in Fig. 1.4.

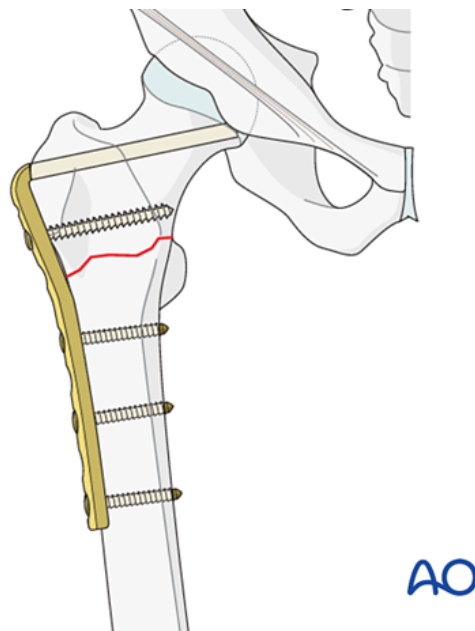


Figure 1.4: Angled Blade Plate

1.4.2 Intramedullary Implants

Intramedullary implants are inserted inside the medullary canal of the femur. Because they lie close to the mechanical axis of the bone, bending stresses are reduced and load transfer becomes more efficient [7]. These implants generally require smaller incisions, less soft tissue stripping, and provide better biomechanical stability in unstable fracture patterns [6].

Common examples include PFN, PFNA-II, and InterTAN.

Proximal Femoral Nail (PFN)

The Proximal Femoral Nail was developed to reduce the bending moment and implant failure risk seen in lateral plate systems [7]. It is inserted through the greater trochanter into the medullary canal and uses proximal screws for fixation of the femoral head and neck.

PFN offers superior load sharing and is commonly preferred for unstable intertrochanteric fractures, reverse oblique fractures, and fractures with subtrochanteric extension [6]. It allows early mobilization and smaller surgical exposure, though technical complications such as screw back-out or improper placement may occur [4].

The Proximal Femoral Nail is shown in Fig. 1.5.

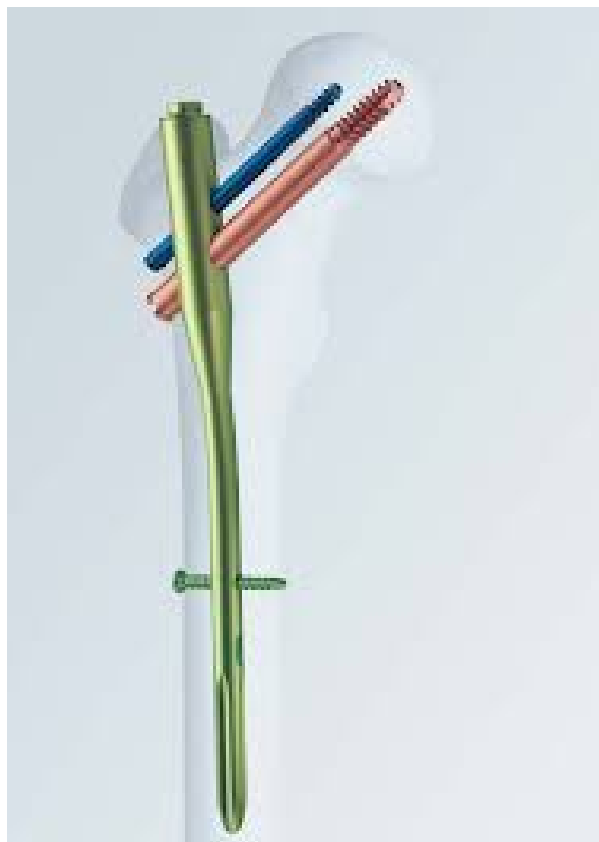


Figure 1.5: Proximal Femoral Nail (PFN)

PFNA-II

PFNA-II is an advanced version of PFN developed especially for osteoporotic bone and anatomical suitability in Asian patients [8]. Instead of a conventional lag screw, it uses a helical blade that compacts cancellous bone during insertion, thereby improving fixation purchase without removing bone material [7].

This design improves fixation strength, increases rotational stability, and reduces the chances of screw cut-out in weak bone [6]. It is therefore commonly preferred in elderly osteoporotic patients with unstable fractures.

The PFNA-II implant is shown in Fig. 1.6.

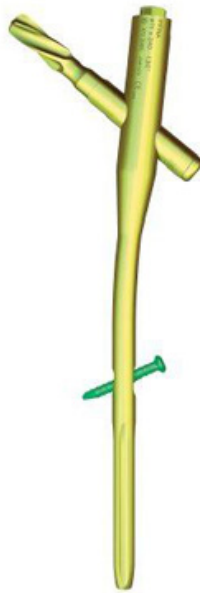


Figure 1.6: PFNA-II

InterTAN

InterTAN is a modern intramedullary implant developed to overcome rotational instability and screw migration problems seen in earlier nail systems [4]. It uses an integrated dual screw mechanism that provides controlled linear compression and strong rotational fixation of the femoral head fragment simultaneously [7].

Due to its high construct stiffness and superior stability, InterTAN is commonly preferred in highly unstable, comminuted, or complex intertrochanteric fractures [6]. However, it is comparatively expensive and surgically more demanding.

The InterTAN implant is shown in Fig. 1.7.

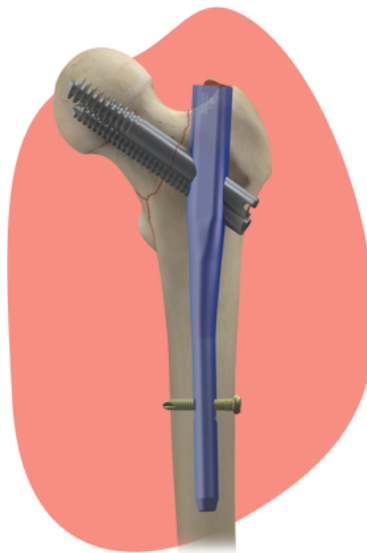


Figure 1.7: InterTAN Nail

1.5 Need for Present Study

Although many implants are clinically available, each design has unique advantages and limitations. Complications such as screw cut-out, implant breakage, excessive sliding, varus collapse, and rotational instability still occur in clinical practice [6, 10].

Therefore, biomechanical comparison using Finite Element Analysis is necessary to evaluate stress distribution, deformation behaviour, and overall stability of different implants under identical loading conditions [3]. This helps in identifying the most suitable implant for intertrochanteric fracture fixation.

1.6 Need for Biomechanical Comparison

Traditionally, implant performance has been evaluated through clinical follow-up studies, cadaveric experiments, or synthetic bone testing [3]. While these methods are useful, they are expensive, time-consuming, and difficult to standardize. In contrast, computational tools provide a faster and more efficient way to study implant behaviour under controlled conditions [11].

Biomechanical comparison helps in understanding [3, 11]:

- Stress concentration in implant and bone
- Deformation under physiological load
- Load-sharing capability
- Construct stiffness
- Probability of implant failure
- Suitability for osteoporotic bone

Such studies assist surgeons in selecting the most suitable implant depending on fracture type and patient condition [7].

1.7 Role of Finite Element Analysis (FEA)

Finite Element Analysis (FEA) is a numerical simulation technique used to study the mechanical response of complex geometries under loading conditions [11]. In FEA, the structure is divided into a large number of small finite elements connected by nodes. Material properties, loads, and constraints are then applied to compute stresses, strains, and deformation [3].

FEA has become an important tool in orthopedic biomechanics because it allows [3, 11]:

- Comparison of multiple implants under identical conditions
- Visualization of internal stress distribution

- Evaluation of bone–implant interaction
- Prediction of weak regions
- Optimization of implant design

Huiskes and Chao [11] pioneered FEA in orthopaedic biomechanics, demonstrating its ability to accurately predict stress distributions in bone and implant constructs. Wirtz et al. [3] provided the foundational experimental material property data for cortical and cancellous femoral bone used to populate such FEA models. In fracture fixation research, FEA therefore provides valuable engineering insight before clinical implementation.

1.8 Scope of Present Work

The present project focuses on the biomechanical comparison of five commonly used implants for intertrochanteric fracture fixation:

- Dynamic Hip Screw (DHS)
- Angled Blade Plate
- Proximal Femoral Nail (PFN)
- PFNA-II
- InterTAN

A three-dimensional CAD model of a healthy femur was developed and modified to simulate an intertrochanteric fracture. Each implant was modelled and assembled with the fractured femur. Material properties of cortical bone, cancellous bone, and titanium alloy were assigned based on published experimental literature [3].

Loading conditions corresponding to one-legged stance of a 75 kg healthy human male were applied [9]. Bergmann et al. [9] established through in-vivo instrumented implant measurements that the hip joint is loaded at approximately 238% of body weight during normal walking, confirming single-leg stance as the most representative physiological loading scenario. Both fractured and healed stages were considered for analysis.

The present study uses simplified assumptions suitable for computational biomechanical comparison. The femur bone was not considered as a rigid body; instead, deformable material behaviour was assigned using isotropic material properties [3]. In reality, human bone exhibits anisotropic and heterogeneous behaviour, where mechanical properties vary significantly with direction and location [3]. Therefore, the current model represents an engineering approximation rather than an exact physiological representation.

Further, detailed simulation of surrounding soft tissues, ligaments, tendons, cartilage, and muscular forces was not incorporated in the present analysis. Because of these simplifications, a perfectly realistic replication of in-vivo loading conditions is beyond the scope of this work.

The study is limited to a healthy adult male femur model and does not specifically address osteoporotic bone conditions, severe bone loss, pathological fractures, or patient-specific anatomical variations. Since intertrochanteric fractures are commonly associated with osteoporosis in elderly patients [6], future studies may incorporate reduced bone density models for more clinically representative analysis.

Despite these limitations, the present work provides a meaningful comparative evaluation of implant biomechanics under controlled and consistent loading conditions, which is valuable for assessing relative implant performance and fixation stability [11].

1.9 Expected Outcome

This study is expected to determine which implant provides more uniform stress distribution and better mechanical stability under simulated loading. Intramedullary implants are generally expected to perform better due to their central position and reduced bending moment arm [6, 7], but the final conclusion depends on comparative FEA results.

The results of this project may contribute to improved implant selection, better fracture healing, reduced revision surgeries, and enhanced patient recovery [4, 8].

Thus, this project combines principles of mechanical engineering, CAD modelling, and biomedical analysis to address an important orthopedic problem through scientific and computational methods.

1.10 Objectives of the Project

Aim: To compare the stability and biomechanical performance of different intertrochanteric fracture fixation implants using Finite Element Analysis.

- To create a CAD model of a healthy femur.
- To develop CAD models of DHS, Angled Blade Plate, PFN, PFNA-II, and InterTAN implants.
- To assemble the femur with each implant.
- To apply appropriate material properties and loading conditions.
- To perform stress analysis on each assembly.
- To identify the implant with maximum stability and mechanical performance.

CHAPTER 2

LITERATURE REVIEW

Intertrochanteric femur fractures are among the most common orthopedic injuries, particularly in elderly patients with osteoporosis [12, 17]. Over the years, several fixation methods such as Dynamic Hip Screw (DHS), Proximal Femoral Nail (PFN), PFNA-II, InterTAN, and blade plate systems have been studied extensively to evaluate their biomechanical stability and clinical performance [13, 18]. The literature available in this field mainly focuses on fixation stability, stress distribution, implant failure mechanisms, and postoperative outcomes [14, 16].

2.1 Biomechanical Comparison of Internal Fixations in Osteoporotic Intertrochanteric Fracture

Gao-Xiang Yuan, Yu-Hui Shen, Bo Chen, and Wei-Bin Zhang [12] conducted a biomechanical comparison of different internal fixation systems for osteoporotic intertrochanteric fractures. Their study emphasized that implant selection significantly affects load transfer, stress concentration, and fixation stability [12]. Intramedullary devices demonstrated better mechanical performance than extramedullary fixation in unstable fracture patterns because of their shorter lever arm and improved load-sharing characteristics [12].

2.2 PFNA-II for Unstable Intertrochanteric Fractures

Vivek Shetty, Vikas Agashe, Sajeev Shekhar, Shaswat Mishra, Kailash M. Jorule, and Suhas Landge [13] studied unstable intertrochanteric femur fracture fixation using PFNA-II. Their clinical study involving 278 patients concluded that PFNA-II provided satisfactory outcomes without the need for additional lateral wall fixation [13]. The helical blade design improved purchase in osteoporotic cancellous bone and reduced rotational instability [13].

2.3 Finite Element Analysis of Implant Selection and Screw Positioning

Tianhao Yang, Faez Noraddin, Baozhe Liu, Zegen Zhang, and Hai-Lun Gu [14] performed finite element analysis for implant selection and screw positioning in proximal femoral basicervical fractures. Their study concluded that implant geometry and screw placement play an important role in reducing stress concentration and micromotion at the fracture site [14]. Proper screw positioning significantly improves fixation stability and decreases risk of cut-out failure [14].

2.4 Biomechanical Comparison of DHS and Gamma Nail Using FEA

Huang et al. [15] conducted a finite element study comparing the Dynamic Hip Screw and Gamma nail for the treatment of unstable trochanteric fractures. Their study demonstrated that the Gamma nail exhibited lower implant stress and better load-sharing properties compared to DHS, particularly in unstable fracture patterns [15]. The results highlighted the biomechanical superiority of intramedullary devices in resisting bending moments and distributing physiological loads more uniformly across the bone–implant construct [15].

2.5 Comparison of Healing Outcomes of Nailing and Plating

Sandeep Rathor, Rashmi Uddanwadiker, Nandram Saryam, and Ashutosh Apte [16] studied the treatment of femur diaphyseal fractures using compression plating and intramedullary nailing through finite element and in-vivo approaches. Their findings indicated that intramedullary nailing generally provided better load sharing and faster healing response, whereas plating systems exhibited higher stress shielding in certain regions [16]. This study also validated the FEA methodology against in-vivo measurements, confirming the reliability of computational biomechanical analysis for evaluating fixation constructs [16].

2.6 PFNA versus DHS for Stable Intertrochanteric Fractures

Mustamsir E, Aji AP, and Fernando A [17] conducted a meta-analysis comparing PFNA with Dynamic Hip Screw for stable intertrochanteric femoral fractures. The study concluded that PFNA offered advantages such as reduced surgical trauma, lower blood loss, and faster post-operative mobilization [17]. However, DHS remained an effective and economical option in stable fracture cases [17].

2.7 PFNA versus DHS for Unstable Intertrochanteric Fractures

Zhang C, Chen Z, Wang M, Chen W, and Ding Z [18] compared clinical outcomes of PFNA and DHS in unstable intertrochanteric femoral fractures. Their meta-analysis concluded that PFNA achieved superior fixation stability, lower complication rates, and improved functional outcomes when compared with DHS in unstable fracture configurations [18].

2.8 Role of Muscle Forces on Strain Distributions in the Proximal Femur

Rogalski, Martelli, Pandy, and Kersh [19] studied the influence of muscle forces on strain distribution in the proximal femur using subject-specific finite element models. Their results showed that inclusion of hip-spanning muscle forces significantly reduced abnormal strain concentrations and changed the loading mode of the femoral neck from bending to compressive

loading. The study concluded that realistic muscle force representation is essential for accurate femur biomechanics simulation [19].

2.9 Positive or Negative Anteromedial Cortical Support of Unstable Pertrochanteric Fractures

Qin Shao, Yue Zhang, Gui-Xin Sun, and co-authors [20] performed a finite element study on unstable pertrochanteric fractures fixed with PFNA-II under different cortical support conditions. They found that positive anteromedial cortical support produced the lowest implant stress and least fracture displacement, whereas negative support generated high stress concentration and instability. Their work highlighted the importance of proper reduction during surgery [20].

2.10 Finite Element Analysis of Four Different Implants for Trochanteric Femoral Fracture

Helwig et al. [21] compared four intramedullary fixation systems inserted in different positions for stabilizing trochanteric femoral fractures. Their analysis showed that both implant design and implant positioning strongly influence stress distribution in the implant and surrounding bone. Caudal positioning of implants provided better mechanical conditions for healing in most cases [21].

2.11 Biomechanical Comparison of Two Femur Bone Implants Using Finite Element Method

Hamidi, Khosravifard, Hematiyan, and co-authors [22] compared Dynamic Hip Screw (DHS) and Gamma Nail for proximal femoral fractures using finite element analysis. Their results indicated that stresses developed in bone were similar for both implants, but implant stresses were consistently higher in DHS. Gamma Nail demonstrated slightly lower fracture displacement and better mechanical reliability [22].

2.12 Trochanteric Nails for Reduction of Intertrochanteric Fractures

Tsai, Lee, Chen, and co-authors [23] carried out a biomechanical study using finite element analysis and Digital Image Correlation for intertrochanteric fractures fixed with trochanteric nails. They reported that closed reduction techniques provided lower stress in the implant and surrounding bone compared to gap reduction. Poor fracture reduction caused higher stresses and increased risk of fixation failure, especially in unstable fractures [23].

2.13 Observations from Literature Survey

Based on the review of available literature, the following common observations were noted:

- Most studies reported that intramedullary implants such as PFN, PFNA-II, Gamma Nail, and InterTAN exhibit superior biomechanical stability in unstable intertrochanteric fractures due to reduced bending moment and improved load sharing [12, 15, 18, 22].
- Dynamic Hip Screw (DHS) remains a reliable, economical, and clinically successful fixation method for stable fracture configurations, especially where lateral wall integrity is maintained [17, 22].
- PFNA-II demonstrated improved fixation in osteoporotic cancellous bone because of its helical blade design, which provides better purchase and rotational stability than conventional lag screws [13, 20].
- Proper implant positioning and screw/blade placement significantly influence stress distribution, construct stiffness, and risk of cut-out failure. Inferior or centrally placed screws were found biomechanically favorable in many studies [14, 21, 23].
- Quality of fracture reduction strongly affects implant performance. Positive cortical support and anatomical reduction conditions resulted in lower implant stress and reduced displacement compared to gap or malreduced fixation [20, 23].
- Implant failure commonly occurs due to screw cut-out, excessive sliding, varus collapse, fatigue fracture, nail breakage, or poor reduction quality [12, 14, 18].
- Finite Element Analysis has become a widely accepted tool for predicting implant biomechanics, comparing fixation systems, and optimizing implant design before clinical application [14–16, 21].
- Several authors highlighted that realistic physiological loading, including muscle forces and gait loading, improves the accuracy of femur stress analysis compared to simplified static loading models [19].
- Most previous studies compared only one or two implants; limited literature is available where multiple commonly used fixation systems are evaluated under identical loading and boundary conditions.
- The majority of available computational studies assumed homogeneous and isotropic bone properties, indicating scope for future patient-specific and osteoporosis-based simulations.

2.14 Limitations Observed in Previous Studies

The literature survey also highlighted several limitations in earlier research:

- Many studies were retrospective in nature.
- Several analyses were based on a single healthy 60-year-old male femur model.

- Patient-specific anatomy and severe osteoporosis were not always considered.
- Static single-leg stance loading was commonly used, while dynamic gait and cyclic loading were neglected.
- Most models assumed isotropic and linear elastic bone properties, whereas actual bone is anisotropic and heterogeneous.

2.15 Research Gap

Although many clinical and computational studies are available, a direct biomechanical comparison of multiple commonly used implants under identical loading, geometry, and material assumptions remains limited. Therefore, the present work focuses on comparing DHS, Angled Blade Plate, PFN, PFNA-II, and InterTAN using Finite Element Analysis under consistent boundary and loading conditions.

CHAPTER 3

MODELLING

3.1 Clinical Consultation

Clinical consultation was carried out with an orthopedic physician to understand the clinical aspects of intertrochanteric fractures, treatment strategies, implant positioning, and practical surgical considerations. Physical implants, particularly screws used in fracture fixation, were obtained from the clinician for dimensional measurements and CAD modelling, as shown in Fig. 3.1.



Figure 3.1: Clinical consultation measurements and dimensional study of implants

3.2 Measurements and Instrumentation

Following the clinical consultation, detailed dimensional measurements of all implant components were carried out to develop accurate three-dimensional CAD models. Since implant geometry directly influences stress distribution, fixation stability, and overall biomechanical performance, obtaining precise measurements was essential. Different measuring instruments were selected based on the size and complexity of the features being measured.

Vernier Caliper

The vernier caliper was used to measure linear dimensions such as implant length, screw diameter, plate thickness, and spacing between holes. Its ability to perform both internal and



Figure 3.2: Vernier Caliper

external measurements made it suitable for general dimensional evaluation. The instrument used for measurement is shown in Fig. 3.2.

Screw Gauge (Micrometer)

The screw gauge was used for precise measurement of small dimensions such as thread diameter and core diameter of screws. Due to its high accuracy, it was suitable for capturing fine geometrical details of threaded components. The instrument is shown in Fig. 3.3.

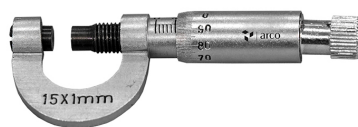
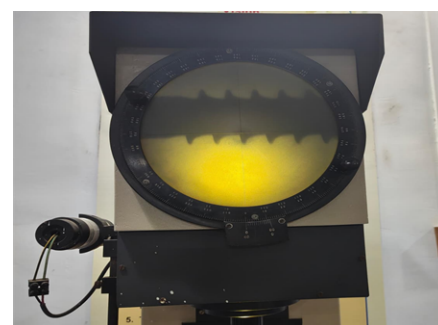


Figure 3.3: Screw Gauge (Micrometer)

Profile Projector

The profile projector was used to measure complex geometrical features such as thread profiles and blade geometry. It enables magnified visualization of the component, allowing accurate measurement of intricate shapes. The instrument is shown in Fig. 3.4.



Helical screw dimensions using profile projector

Figure 3.4: Profile Projector

Digital Angle Protractor

The digital angle protractor was used to measure angular parameters such as the neck-shaft angle, screw inclination, and blade orientation. Accurate angular measurements are important

for proper implant alignment. The instrument is shown in Fig. 3.5.



Figure 3.5: Digital Angle Protractor

Data Utilization

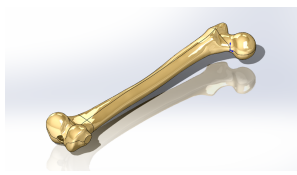
All measured dimensions were systematically recorded and used to develop precise CAD models in SolidWorks. These models were subsequently imported into ANSYS for finite element analysis. The accuracy of measurements ensured realistic simulation results.

3.3 Femur Fracture Modeling

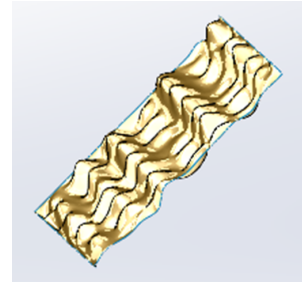
A three-dimensional CAD model of a healthy femur was developed using anatomical references corresponding to a healthy adult male of approximately 75 kg body weight, as shown in Fig. 3.6. The model included the proximal, shaft, and distal regions of the femur, ensuring anatomical accuracy for subsequent analysis.

An anatomically realistic intertrochanteric fracture was created based on orthopedic consultation and X-ray references. A fracture plane was generated and applied to the femur model (Fig. 3.6), which was then used to split the bone into two distinct fragments. The fractured configuration of the femur and the resulting separation of fragments are illustrated in Fig. 3.7.

The final CAD models of both the healthy and fractured femur, including multiple views for better visualization, are presented in Fig. 3.8. These models were further utilized for implant assembly and finite element analysis.



Normal femur model



Fracture plane generation

Figure 3.6: Creation of intertrochanteric fracture plane on femur model



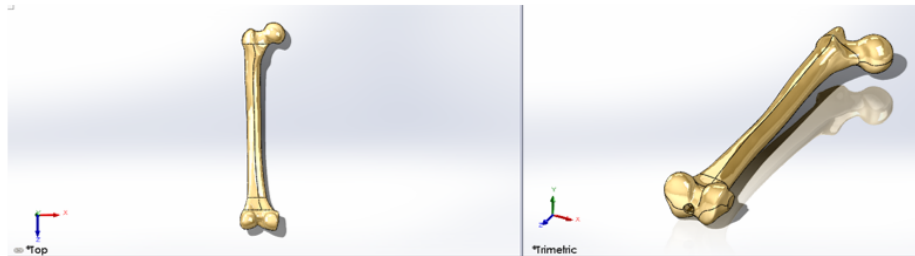
Femur with fracture plane



Separated fragments

Figure 3.7: Fractured femur model after split operation

Normal Healthy Bone



Fractured Bone

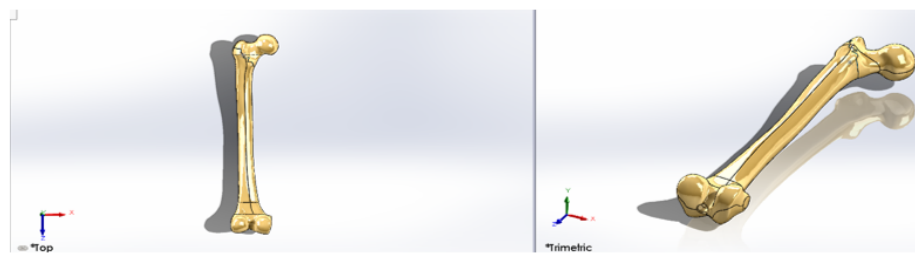


Figure 3.8: Final CAD model views of healthy and fractured femur

3.4 Implant Modeling and Assembling

Detailed CAD models of all implants were created in SolidWorks based on measured dimensions and manufacturer references. Each implant was assembled individually with the fractured femur in clinically correct orientation.

3.4.1 Dynamic Hip Screw (DHS)

DHS provides controlled sliding and compression at the fracture site, which promotes fracture healing. The Dynamic Hip Screw model and its assembly with the femur are shown in Fig. 3.9.

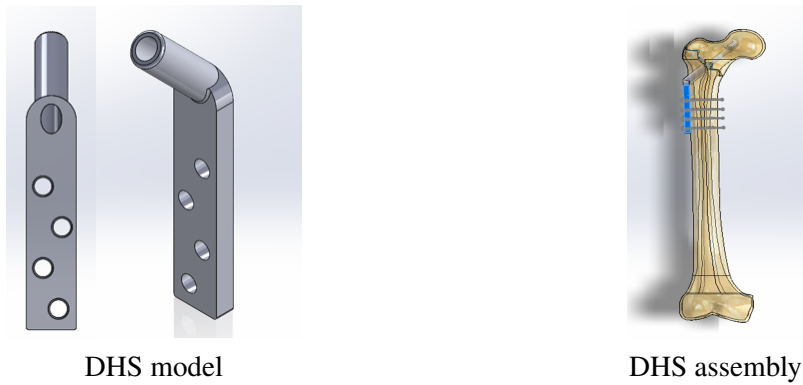


Figure 3.9: Dynamic Hip Screw implant and femur assembly

3.4.2 Angled Blade Plate

The angled blade plate provides rigid angular fixation and maintains alignment in unstable fracture conditions. The Angled Blade Plate model and its assembly with the femur are shown in Fig. 3.10.

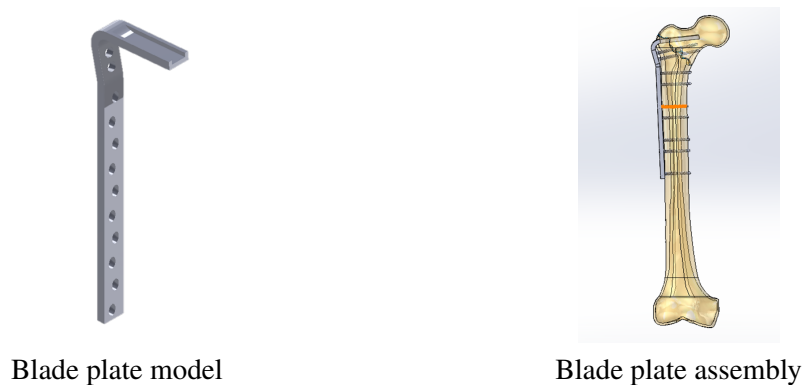


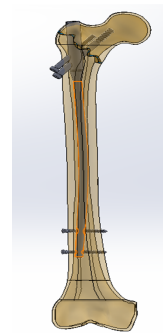
Figure 3.10: Angled Blade Plate implant and femur assembly

3.4.3 Proximal Femoral Nail (PFN)

PFN offers superior load sharing due to its intramedullary position and reduced bending stresses. The PFN model and its assembly with the femur are shown in Fig. 3.11.



PFN model



PFN assembly

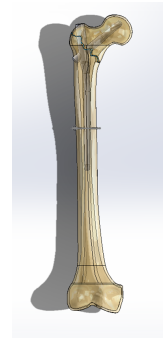
Figure 3.11: Proximal Femoral Nail implant and femur assembly

3.4.4 PFNA-II

PFNA-II uses a helical blade which provides improved purchase in cancellous bone and better rotational stability. The PFNA-II model and its assembly with the femur are shown in Fig. 3.12.



PFNA-II model



PFNA-II assembly

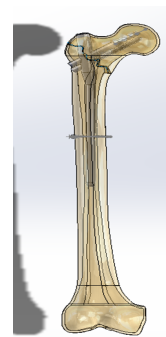
Figure 3.12: PFNA-II implant and femur assembly

3.4.5 InterTAN

InterTAN consists of a dual integrated screw mechanism that improves axial and rotational stability. The InterTAN model and its assembly with the femur are shown in Fig. 3.13.



InterTAN model



InterTAN assembly

Figure 3.13: InterTAN implant and femur assembly

CHAPTER 4

ANALYSIS

The femur model was divided into cortical bone and cancellous bone regions. Material properties were assigned based on literature values. All materials were assumed to be homogeneous, isotropic, and linearly elastic and Finite Element Analysis was carried out for all cases. Titanium alloy (Ti-6Al-4V) properties were assigned to all implant components.

4.1 Material Properties

The femur bone model was divided into cortical and cancellous regions. Each region was assigned separate material properties derived from literature so that the stiffness contrast between the dense outer cortex and the inner trabecular bone is captured accurately in the analysis. All materials were assumed to be homogeneous, isotropic, and linearly elastic. Physiological loading conditions representing a one-legged stance and daily activities were applied at the femoral head, with distal constraints at the condyles.

The material properties used in the finite element simulations were assigned based on values available in literature. All materials were assumed to be linearly elastic and isotropic for simplification of the numerical model. The properties assigned to cortical bone, cancellous bone, and titanium alloy (Ti-6Al-4V) are listed below.

Properties of Cortical Bone

These properties are taken from the ANSYS Granta Edupack. [24]

Table 4.1: Mechanical Properties of Cortical Bone

Property	Value
Density	1800 kg/m ³
Young's Modulus	18 GPa
Poisson's Ratio	0.16 – 0.30
Tensile Yield Strength	135 MPa
Tensile Ultimate Strength	167 MPa
Compressive Yield Strength	130 MPa
Compressive Ultimate Strength	250 MPa

Properties of Cancellous Bone

Properties of Titanium Alloy (Ti-6Al-4V)

These properties are taken directly from the "ANSYS workbench engineering database".

Table 4.2: Mechanical Properties of Cancellous Bone

Property	Value
Density	350 kg/m ³
Young's Modulus	3 GPa
Poisson's Ratio	0.30 – 0.35
Tensile Yield Strength	30 MPa
Compressive Ultimate Strength	12 MPa

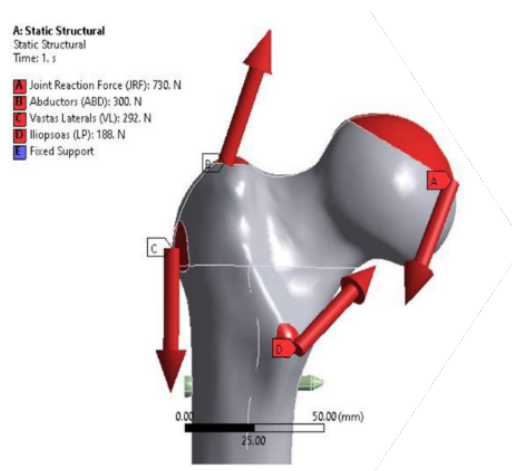
Table 4.3: Mechanical Properties of Titanium Alloy (Ti-6Al-4V)

Property	Value
Density	4429 kg/m ³
Young's Modulus	111.2 GPa
Poisson's Ratio	0.3387
Tensile Yield Strength	860 MPa
Tensile Ultimate Strength	950 MPa

4.2 Loading Conditions

Physiological loading conditions were applied to simulate the actual forces acting on the femur during daily activities. The loading conditions selected for the present study were based on body weight transfer through the hip joint during common activities such as standing and stair climbing. These loading cases are important because they generate significant stresses in the proximal femur and are commonly considered in biomechanical studies.

For the present analysis, a healthy male subject of approximately **75 kg body weight** was considered. The corresponding external loads were calculated based on gravitational force and distributed to represent joint reaction forces and muscle forces acting during one-legged stance as shown in Fig. 4.1.

**Figure 4.1: Loading directions and force components considered for femur analysis**

4.2.1 One-Legged Stance Loading

One-legged stance is one of the most critical loading conditions for the femur because the entire body weight is transferred through a single lower limb during walking. Under this condition, the hip experiences high compressive and bending loads due to body weight and abductor muscle action as shown in Fig. 4.2.

This loading case was selected to evaluate the stress distribution and deformation behaviour of the intact femur as well as implant-fixed fractured femur models.

Muscle and joint reaction forces	Magnitude of resultant force (N)	Orientation (degree)	
		φ	γ
Joint reaction force (JRF)	730	291	273
Abductors (ABD)	300	110	90
Vastus lateralis (VL)	292	270	270
Iliopsoas (LP)	188	99	137

Figure 4.2: One-Legged Stance Loading Conditions for a Human Male of 75 kg

4.2.2 Reduced Initial Healing Load

During the early postoperative healing phase, patients are generally advised partial weight bearing to avoid excessive stress at the fracture site. To represent this condition, the applied loading values were reduced to approximately one-tenth of the full physiological load.

This reduced loading condition was used to simulate the initial healing stage and compare implant behaviour under safer postoperative conditions.

4.2.3 Boundary Conditions

The distal end of the femur was assigned fixed support in the finite element model to restrict rigid body motion and replicate support provided by the lower limb during stance. External forces were applied at the femoral head and greater trochanter region according to the selected loading case.

These loading and boundary conditions were kept consistent for all implant assemblies to ensure a fair biomechanical comparison.

4.3 Finite Element Analysis

Fig. 4.3 shows Finite Element Analysis (FEA) that was carried out to evaluate the stress distribution and deformation behaviour of the femur under physiological loading and with different implant configurations. A baseline analysis was first performed on a normal, healthy femur without any fracture or implant in order to validate the loading conditions and boundary constraints. The same meshing strategy, material properties, and loading will be subsequently used for all implant-based models.

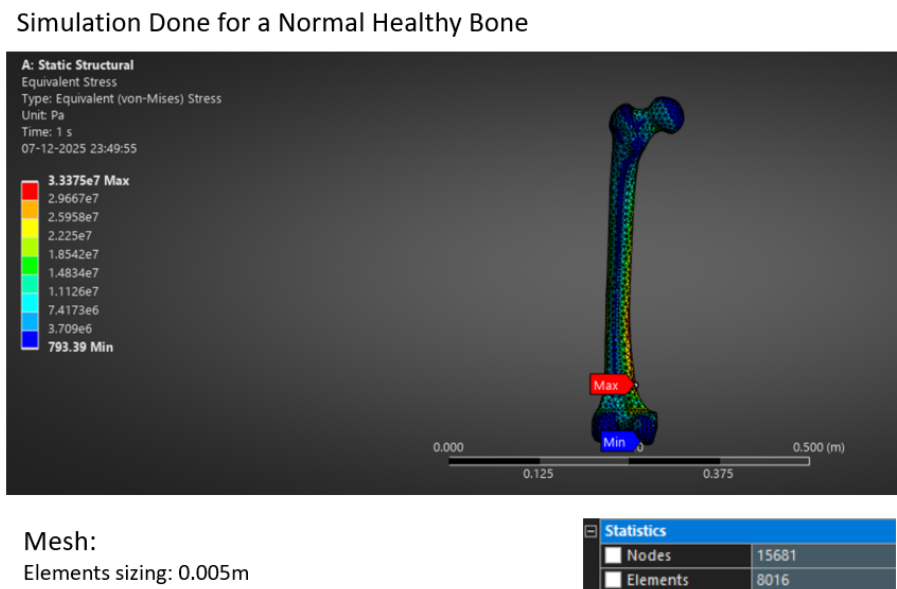


Figure 4.3: Finite Element stress distribution on a normal healthy femur under one-legged stance loading

4.3.1 Contact Conditions and Boundary Supports

For all implant assemblies, identical loading and support philosophy was maintained to ensure a fair biomechanical comparison. The distal end of the femur was assigned fixed support to restrict rigid body motion and simulate lower limb support during one-legged stance. Physiological loads corresponding to a 75 kg body weight and reduced 7.5 kg loading conditions were applied at the femoral head region.

Dynamic Hip Screw (DHS)

- No separation between cortical screws and bone
- Rough contact between upper and lower fracture fragments for fractured model
- Bonded contact between fracture fragments for healed model

- Bonded contact between lag screw and DHS plate
- Frictionless contact between sliding screw and DHS barrel
- No separation between DHS plate and bone
- Contacts shown visually in Fig. 4.4

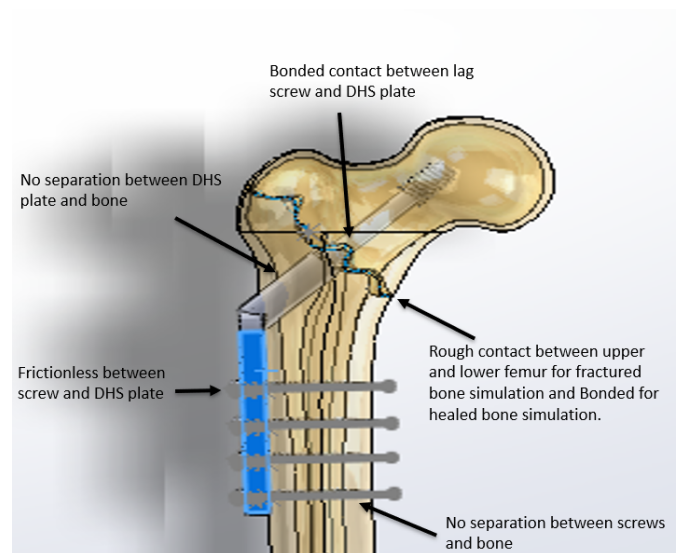


Figure 4.4: Contact conditions used for DHS assembly

Angled Blade Plate

- No separation between fixation screws and bone
- Rough contact between fracture fragments for fractured model
- Bonded contact between fracture fragments for healed model
- Frictionless contact between screw and blade plate
- No separation between angled blade plate and bone
- Contacts shown visually in Fig. 4.5

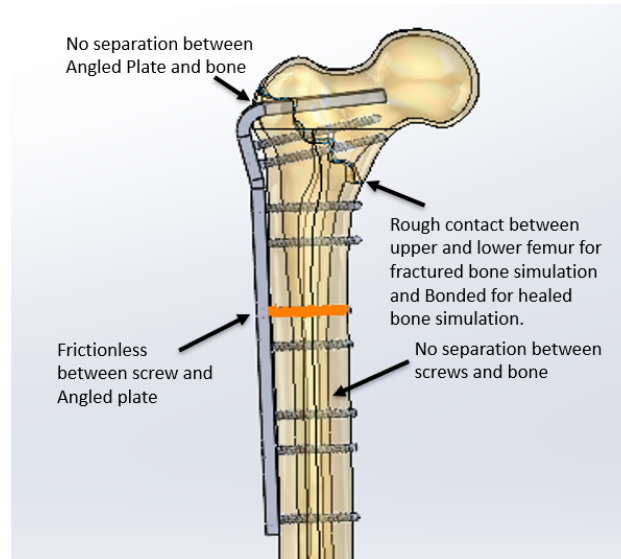


Figure 4.5: Contact conditions used for Angled Blade Plate assembly

Proximal Femoral Nail (PFN)

- No separation between nail and bone canal
- No separation between proximal screws and bone
- Rough contact between fracture fragments for fractured model
- Bonded contact between fracture fragments for healed model
- Frictionless contact between screw and PFN nail
- Contacts shown visually in Fig. 4.6

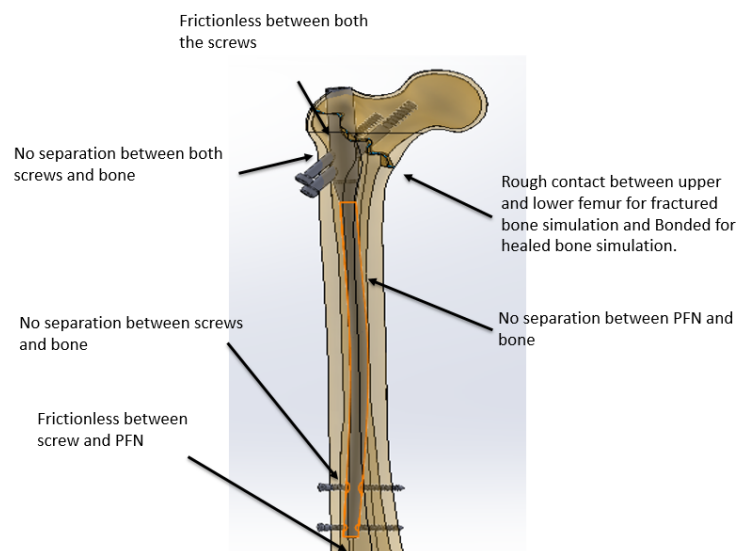


Figure 4.6: Contact conditions used for PFN assembly

PFNA-II

- No separation between PFNA-II nail and bone
- Rough contact between fracture fragments for fractured model
- Bonded contact between fracture fragments for healed model
- No separation between screws and bone
- No separation between helical blade and bone
- Frictionless contact between screw and PFNA-II nail
- Frictionless contact between helical blade and PFNA-II nail
- Contacts shown visually in Fig. 4.7

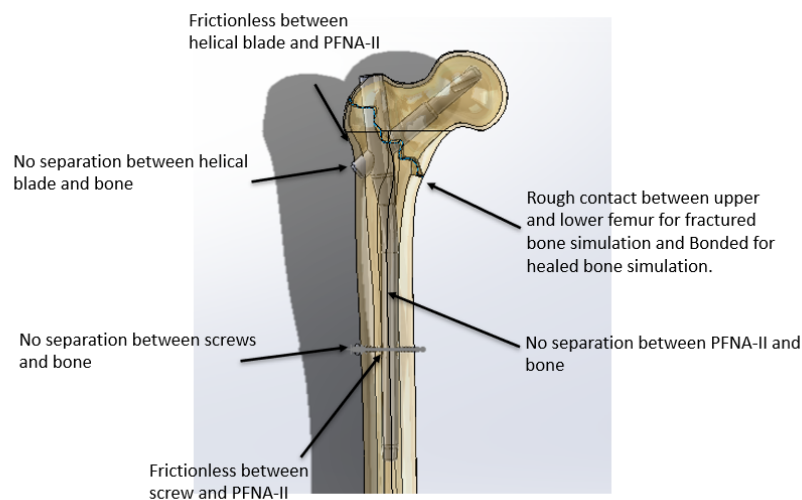


Figure 4.7: Contact conditions used for PFNA-II assembly

InterTAN

- No separation between InterTAN nail and bone
- Rough contact between fracture fragments for fractured model
- Bonded contact between fracture fragments for healed model
- No separation between screws and bone
- No separation between integrated dual screws and bone
- Frictionless contact between screw and InterTAN nail

- Frictionless contact between dual integrated screws
- Contacts shown visually in Fig. 4.8

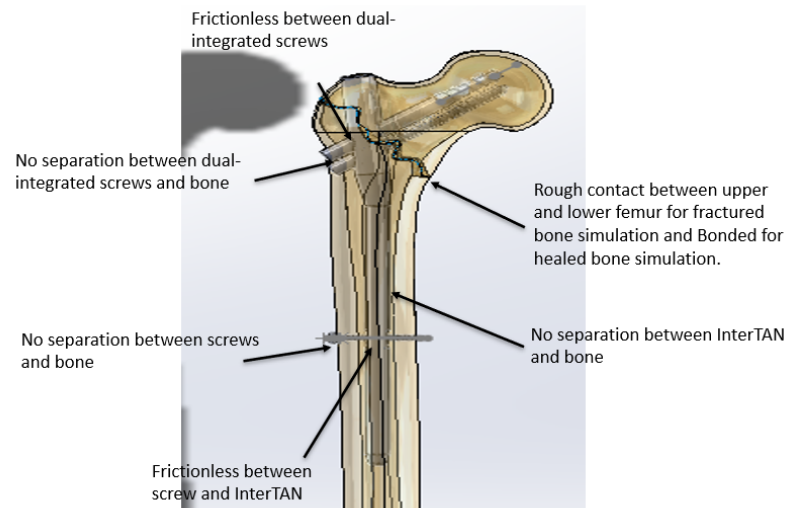


Figure 4.8: Contact conditions used for InterTAN assembly

CHAPTER 5

RESULTS AND DISCUSSION

Finite Element Analysis was performed on all implant assemblies under fractured, healed (bonded), and reduced loading conditions wherever applicable. The results were evaluated in terms of maximum equivalent (Von Mises) stress in bone, maximum stress in implant and average stress distribution. A healthy femur model was also analysed as a reference for comparison.

All implant systems were compared on the basis of stress concentration, deformation behaviour, load-sharing efficiency, and overall biomechanical stability to identify the most suitable implant for intertrochanteric fracture fixation.

The analysis indicates that, the maximum stress observed in the healed bone is generally lower than that in the normal healthy bone model. This reduction can be attributed to stress redistribution caused by the presence of the implant, which alters the load path and reduces the bending moment experienced by the cortical bone. The phenomenon is consistent with partial stress shielding, where the implant absorbs a considerable share of the load, thereby decreasing stress within the surrounding bone. The maximum stress in the healthy bone was 33.75 MPa overall; however, in the region corresponding to peak stress in the implant configurations, it was 17.39 MPa, while the highest stresses in those cases were predominantly observed within the implants.

Additionally, the absence of soft tissue modelling (muscles, ligaments, and joint contact forces) may contribute to this difference, as real physiological loading conditions involve complex load sharing through soft tissues that influence stress distribution. In the healthy bone model, the maximum stress was observed near the mid-shaft region, primarily due to bending moments generated by the applied load acting away from the neutral axis. However, in the implanted models, this bending effect is reduced since the implant provides an alternate load-bearing path, thereby minimizing stress concentration in the shaft region. These observations highlight the significant role of implants in modifying biomechanical behaviour and emphasize the importance of balanced load sharing to avoid excessive stress shielding and maintain bone integrity.

5.1 Results of Dynamic Hip Screw (DHS)

As shown in Fig. 5.1, the fractured femur fixed with DHS under a 75 kg load showed a maximum equivalent stress of **638.29 MPa** in the implant. The bone experienced a peak stress of **1.36 GPa**, concentrated near the cortical region close to the fracture site, indicating a potential risk of localized failure under high loading.

Under a reduced load of 7.5 kg, the average stress in the implant was **1.34 MPa**, while the

bone experienced a maximum stress of **216.7 MPa**.

The average stress observed in the DHS model under full loading was **4.39 MPa**.

For the bonded condition under 75 kg loading, the implant showed a maximum stress of **789.24 MPa**, while the average stress was **5.80 MPa**. The corresponding literature value reported for DHS was approximately **512 MPa** [15].

The results indicate that DHS provides stable fixation; however, due to its extramedullary configuration, higher bending stresses are induced in the system compared to intramedullary devices.

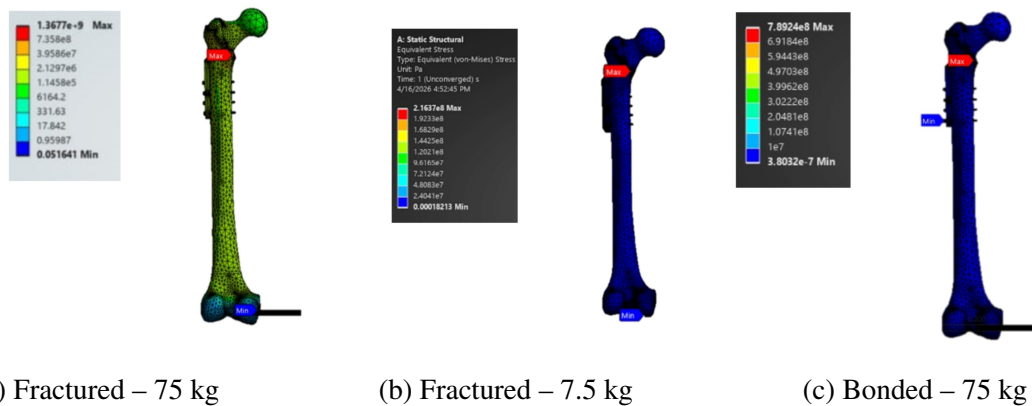


Figure 5.1: Finite Element Analysis results of DHS implant under different loading and healing conditions

5.2 Results of Angled Blade Plate

As shown in Fig. 5.2, the fractured femur fixed with the Angled Blade Plate under a 75 kg load showed a maximum equivalent stress of **807.79 MPa**, with peak stress observed in the implant near the proximal fixation region. The corresponding stress in the bone reached **226.09 MPa**, concentrated near the fracture interface.

Under reduced loading of 7.5 kg, the implant stress decreased to **242.5 MPa**, while the bone experienced a maximum stress of **64.53 MPa**. This indicates a proportional reduction in load transfer under lower loading conditions.

In the bonded condition at 75 kg, the bone stress reduced significantly to **30.97 MPa**, indicating improved load sharing after fracture healing. The average stress in the implant system was observed to be **4.09 MPa** under fractured loading and **6.65 MPa** under bonded conditions.

These results suggest that the Angled Blade Plate provides rigid fixation; however, higher stresses in the implant indicate increased bending effects due to its extramedullary nature.

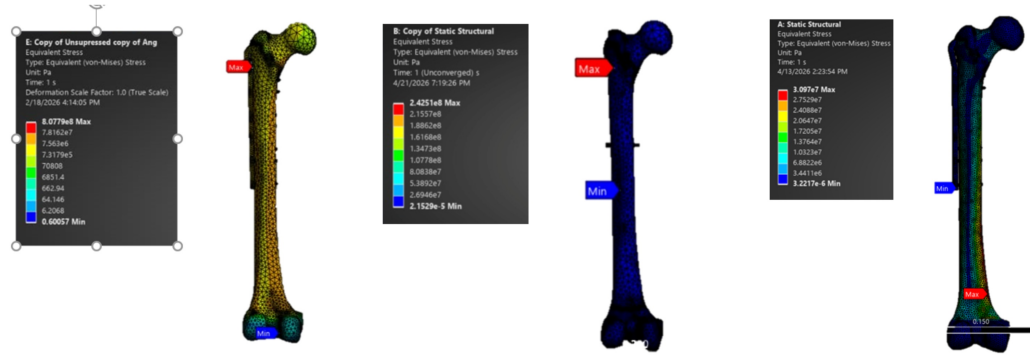


Figure 5.2: Finite Element Analysis results of Angled Blade Plate implant under different loading and healing conditions

5.3 Results of Proximal Femoral Nail (PFN)

As shown in Fig. 5.3, the fractured femur fixed with PFN under a 75 kg load showed a maximum implant stress of **1.84 GPa**, while the bone experienced a peak stress of **760.11 MPa**, indicating significant load transfer through both components.

Under reduced loading of 7.5 kg, the implant stress decreased to **453.5 MPa**, and the bone stress reduced to **122.62 MPa**.

In the bonded condition under 75 kg loading, the implant stress was **663.32 MPa**, while the bone stress reduced significantly to **37.33 MPa**, showing improved load distribution after healing. The average stresses were **11.10 MPa**, **3.09 MPa**, and **6.74 MPa** for fractured 75 kg, fractured 7.5 kg, and bonded conditions respectively.

These results indicate that PFN provides better load-sharing characteristics due to its intramedullary placement, resulting in reduced bending moments compared to extramedullary implants.

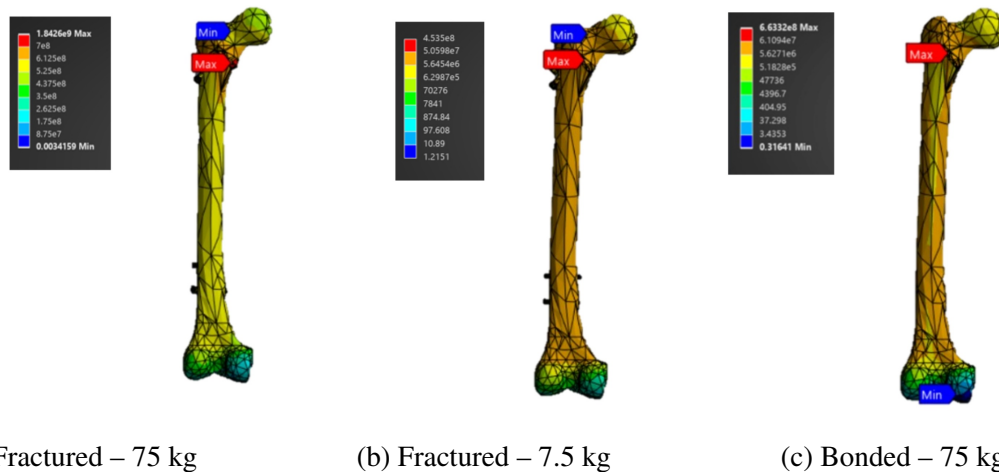


Figure 5.3: Finite Element Analysis results of PFN implant under different loading and healing conditions

5.4 Results of PFNA-II

As shown in Fig. 5.4, the fractured femur fixed with PFNA-II under a 75 kg load showed a maximum implant stress of **1.04 GPa**, while the bone experienced a peak stress of **2.1481 GPa**, indicating high localized stress near the fracture region.

Under reduced loading of 7.5 kg, the implant stress reduced to **242.51 MPa**, and the bone stress was **167.74 MPa**.

The average stress observed under fractured loading was **13.21 MPa** and **1.17 MPa** under reduced loading. The higher stress values in bone are attributed to the helical blade design, which compacts cancellous bone and provides strong fixation.

For the bonded condition under 75 kg loading, the implant stress reduced significantly to **131.04 MPa**, while the bone stress was **29.48 MPa**, indicating improved load transfer after healing.

Overall, PFNA-II demonstrates improved fixation stability, especially in osteoporotic conditions, although localized stress concentrations may be significant.

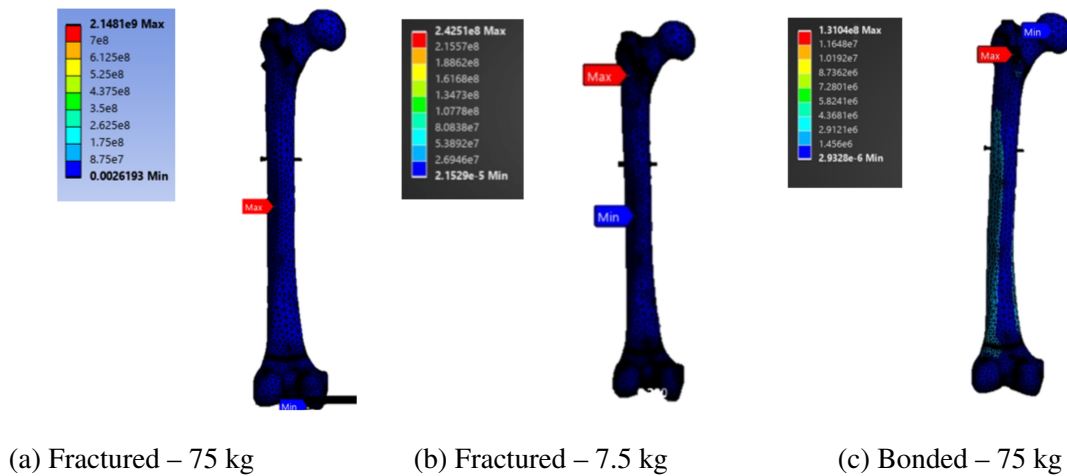


Figure 5.4: Finite Element Analysis results of PFNA-II implant under different loading and healing conditions

5.5 Results of InterTAN

As shown in Fig. 5.5, the fractured femur fixed with InterTAN under a 75 kg load showed a maximum implant stress of **2.6169 GPa**, indicating high stiffness of the construct. The average stress under this condition was **17.45 MPa**.

Under reduced loading of 7.5 kg, the implant stress decreased to **215.39 MPa**, while the bone experienced a peak stress of **62.26 MPa**. The corresponding average stress was **1.66 MPa**.

In the bonded condition under 75 kg loading, the implant stress was **90.99 MPa**, while the bone stress reduced to **31.65 MPa**. The average stress in this condition was **7.67 MPa**, indicating improved load transfer after healing.

These results suggest that InterTAN provides superior stability due to its dual screw mechanism, which enhances rotational stability and load sharing, making it highly suitable for unstable intertrochanteric fractures.

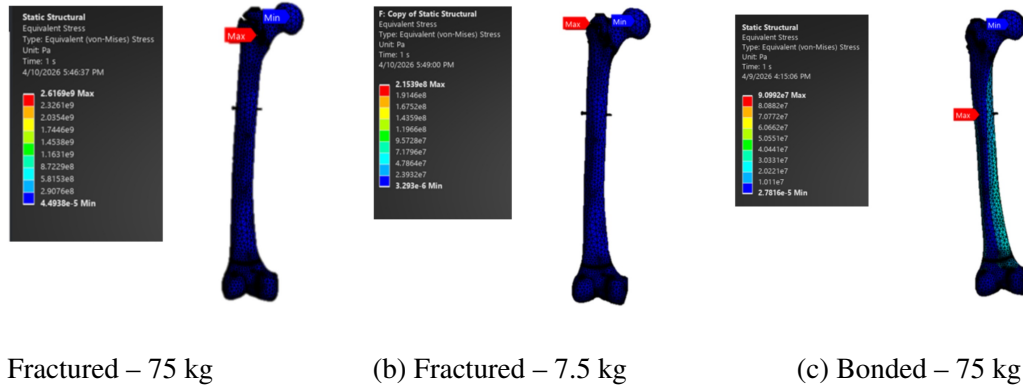


Figure 5.5: Finite Element Analysis results of InterTAN implant under different loading and healing conditions

5.6 Overall Comparison

Among extramedullary implants, Angled Blade Plate performed better than the DHS with respect to stress. Among intramedullary devices, PFN and InterTAN exhibited better load-sharing characteristics by having similar average stresses in healed simulations compared to normal healthy bone.

Overall, intramedullary implants demonstrated mechanical behaviour closer to the healthy femur model and are more suitable for unstable intertrochanteric fracture fixation. All the cases shown below 5.6

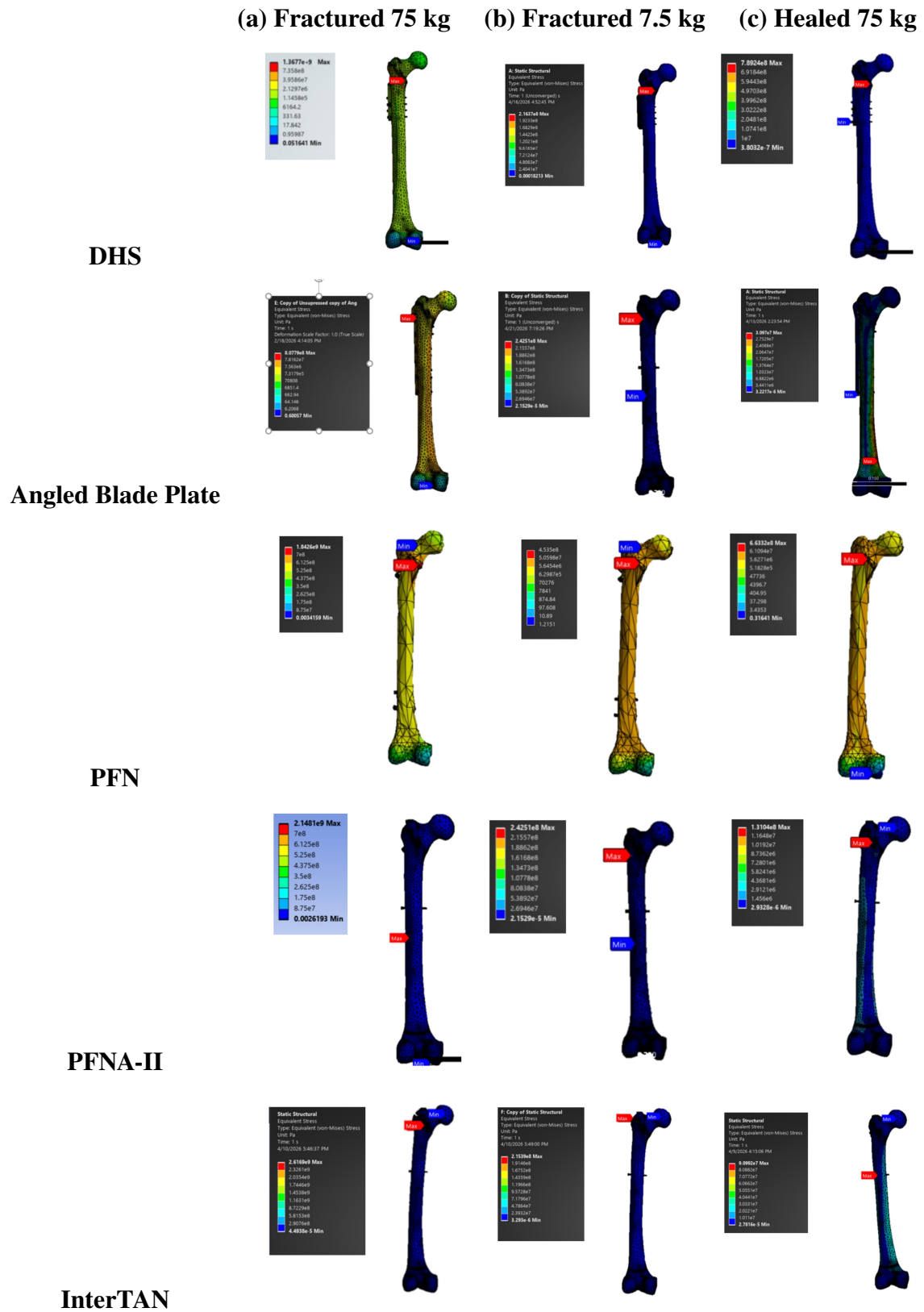


Figure 5.6: Comparison of stress distribution for different implants under fractured (75 kg), reduced (7.5 kg), and healed (75 kg) loading conditions

Table 5.1: Overall Finite Element Analysis Results with Literature Comparison

Implant	Condition	Location	Max Stress	Average Stress	Literature Stress
Healthy Bone	–	Bone	33.75 MPa	8.23 MPa	–
Angled Blade Plate	Fractured 75 kg	Implant	807.79 MPa	4.09 MPa	–
		Bone	226.09 MPa		–
	Fractured 7.5 kg	Implant	242.5 MPa	1.16 MPa	–
		Bone	64.53 MPa		–
	Bonded 75 kg	Implant		6.65 MPa	–
		Bone	30.97 MPa		–
DHS Plate	Fractured 75 kg	Implant	638.29 MPa	4.39 MPa	–
		Bone	1.36 GPa		–
	Fractured 7.5 kg	Implant		1.34 MPa	–
		Bone	216.7 MPa		–
	Bonded 75 kg	Implant	789.24 MPa	5.80 MPa	512 MPa [15]
		Bone	26.03 MPa		–
PFN	Fractured 75 kg	Implant	1.84 GPa	11.10 MPa	–
		Bone	760.11 MPa		–
	Fractured 7.5 kg	Implant	453.5 MPa	3.09 MPa	–
		Bone	122.62 MPa		–
	Bonded 75 kg	Implant	663.32 MPa	6.74 MPa	858 MPa (2800 N) [25]
		Bone	37.33 MPa		–
PFNA-II	Fractured 75 kg	Implant	1.04 GPa	13.21 MPa	–
		Bone	2.1481 GPa		–
	Fractured 7.5 kg	Implant	242.51 MPa	1.17 MPa	–
		Bone	167.74 MPa		–
	Bonded 75 kg	Implant	131.04 MPa	10.02 MPa	515 MPa [14]
		Bone	29.48 MPa		–
InterTAN	Fractured 75 kg	Implant	2.6169 GPa	17.45 MPa	–
		Bone			–
	Fractured 7.5 kg	Implant	215.39 MPa	1.66 MPa	–
		Bone	62.26 MPa		–
	Bonded 75 kg	Implant	90.99 MPa	7.67 MPa	182.4 MPa (2300 N) [14]
		Bone	31.65 MPa		–

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

In this study, a comprehensive biomechanical comparison of commonly used implants for intertrochanteric femur fracture fixation was carried out using finite element analysis. A three-dimensional CAD model of the femur was developed and modified to simulate an intertrochanteric fracture based on clinical inputs. Five implants, namely Dynamic Hip Screw (DHS), Angled Blade Plate, Proximal Femoral Nail (PFN), PFNA-II, and InterTAN, were modelled and analysed under identical loading and boundary conditions.

The results demonstrated that implant design significantly influences stress distribution, deformation, and overall stability of the fixation system. The maximum stress in the healthy bone was 33.75 MPa overall; however, in the region corresponding to peak stress in the implant configurations, it was 17.39 MPa, while the highest stresses in those cases were predominantly observed within the implants. Intramedullary implants such as PFN, PFNA-II, and InterTAN showed comparatively better load-sharing characteristics due to their alignment with the mechanical axis of the femur. Among all the implants, InterTAN exhibited improved stability under both fractured and reduced loading conditions, while PFNA-II provided effective fixation due to its helical blade design.

Extramedullary implants such as DHS and Angled Blade Plate showed higher stress concentration in certain regions, indicating comparatively lower load-sharing efficiency. However, they remain suitable for stable fracture conditions due to their simplicity and clinical reliability.

The simulation results support the clinical perspective that it is important to note that full body weight is not applied immediately after implant fixation. Applying complete physiological loading on a freshly fractured femur may lead to excessive stress at the fracture site, potentially worsening the fracture or causing implant failure. The stresses in all implants' simulations approached or exceeded the tensile strength of bone in localized regions. This indicates a high risk of failure if full body weight is applied prematurely. Reduced loading conditions showed comparatively lower stress levels, reinforcing the importance of gradual load progression during recovery. Therefore, a controlled rehabilitation period is recommended, during which patients use external support such as walkers or crutches to limit load transfer until sufficient healing occurs.

Among all the implants analysed, InterTAN demonstrated performance closest to that of the healthy femur model in terms of average stress distribution in healed bone case. This indicates superior load-sharing capability, which is a critical factor in fracture healing. Effective load sharing ensures that the bone continues to experience physiological stresses, thereby preventing stress shielding. If the implant carries a majority of the load, the surrounding bone may

become osteoporotic over time due to reduced mechanical stimulation. Hence, InterTAN can be considered a more favorable option for maintaining biomechanical compatibility and promoting long-term bone health.

Overall, the study highlights the importance of implant selection based on fracture type, loading conditions, and biomechanical performance. Finite element analysis proved to be an effective tool for evaluating implant behaviour and understanding stress distribution patterns in fracture fixation systems.

6.2 Future Scope

Although the present study provides valuable insights into the biomechanical performance of different implants, certain limitations exist which can be addressed in future work.

The current analysis assumes isotropic and homogeneous material properties for bone, which does not fully represent the complex anisotropic nature of real human bone. Future studies can incorporate anisotropic and heterogeneous material models for more realistic simulations.

The model used in this study represents a healthy adult male femur and does not consider osteoporotic bone conditions, which are highly relevant in intertrochanteric fractures. Including osteoporotic bone properties would provide a more clinically accurate comparison of implant performance.

Soft tissues, muscles, and ligaments were not included in the simulation. Their inclusion can improve the accuracy of load transfer and boundary conditions, leading to more realistic results.

The study is limited to static loading conditions. Future work can include dynamic and cyclic loading analysis to simulate real-life activities such as walking and stair climbing, as well as fatigue behaviour of implants.

Further research can also explore optimization of implant design parameters, such as screw positioning, blade geometry, and material selection. Additionally, patient-specific modelling using CT scan data can be incorporated to develop customized implant solutions.

Overall, future advancements in modelling techniques and computational methods can significantly enhance the accuracy and clinical relevance of biomechanical studies in orthopedic implant design.

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ANNEXURE 1

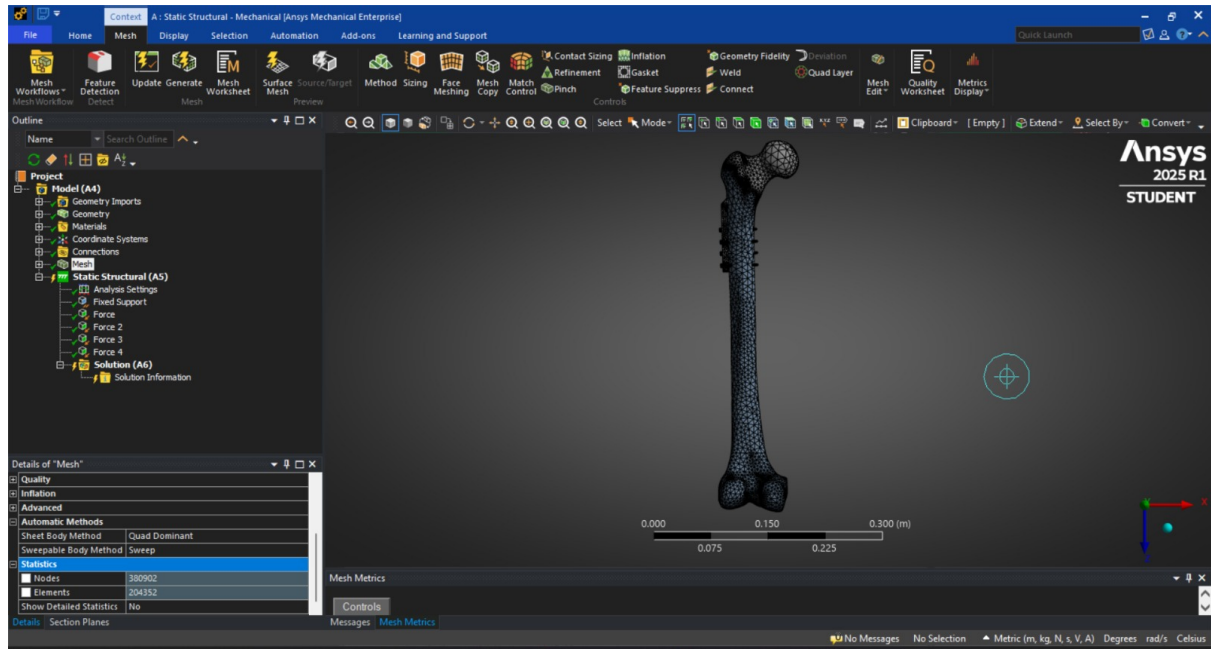


Figure 6.1: Finite Element Mesh of Dynamic Hip Screw (DHS) Implant Model

ANNEXURE 2

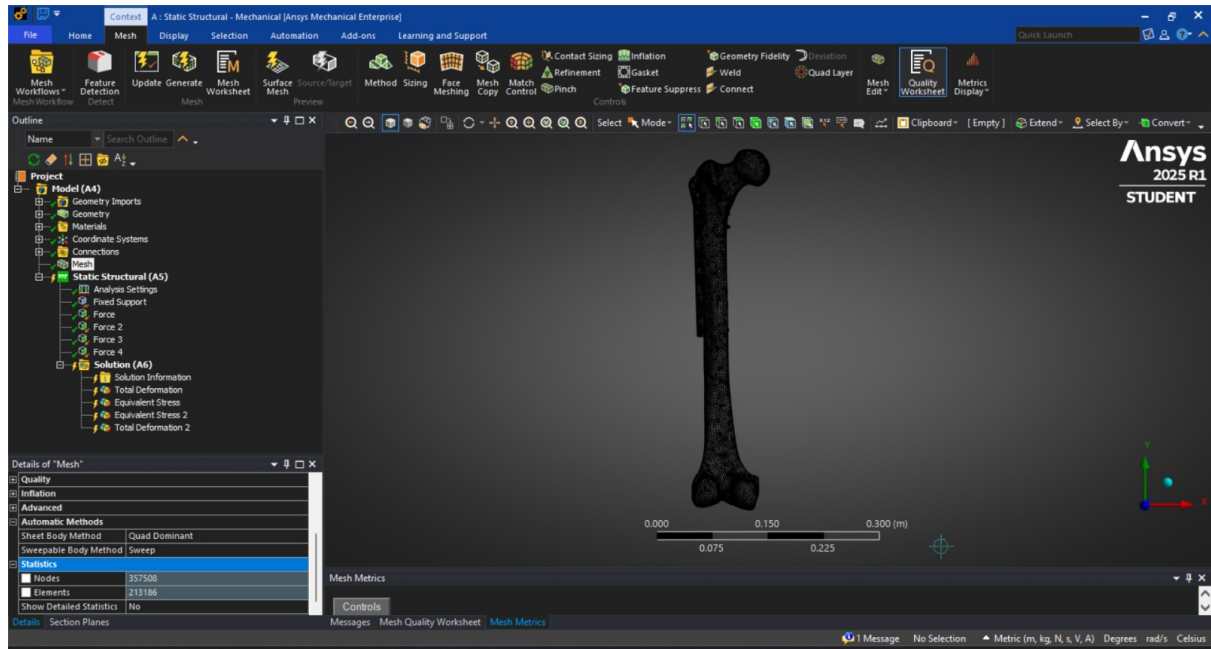


Figure 6.2: Finite Element Mesh of Angled Blade Plate Implant Model

ANNEXURE 3

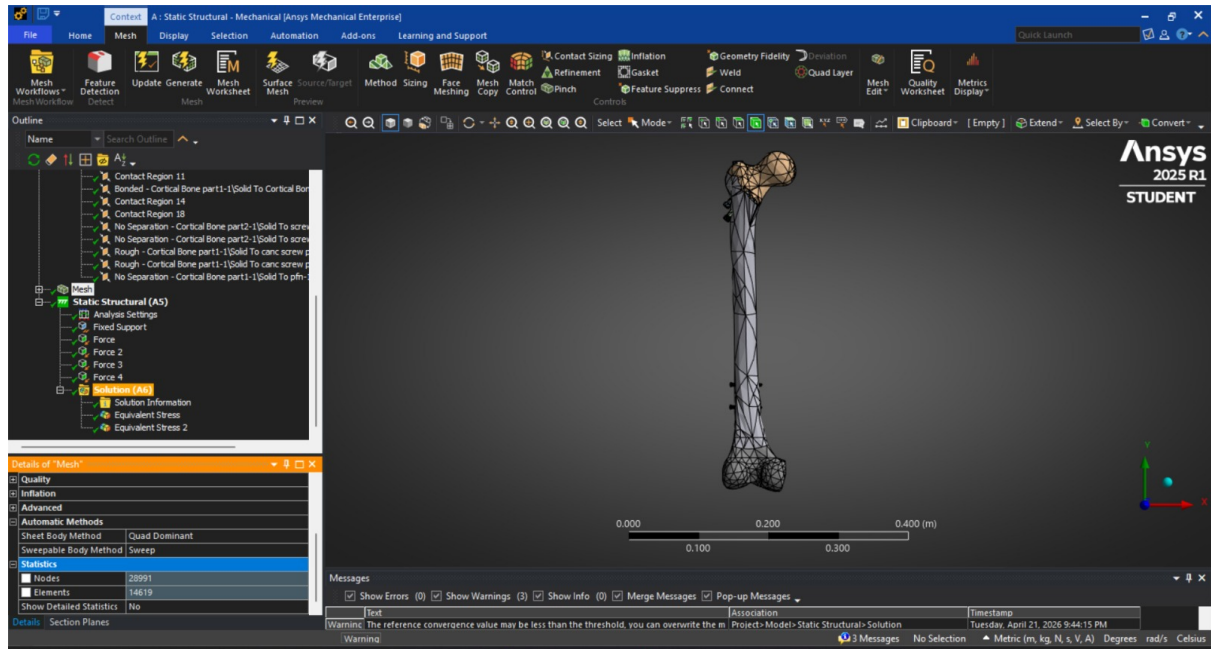


Figure 6.3: Finite Element Mesh of Proximal Femoral Nail (PFN) Implant Model

ANNEXURE 4

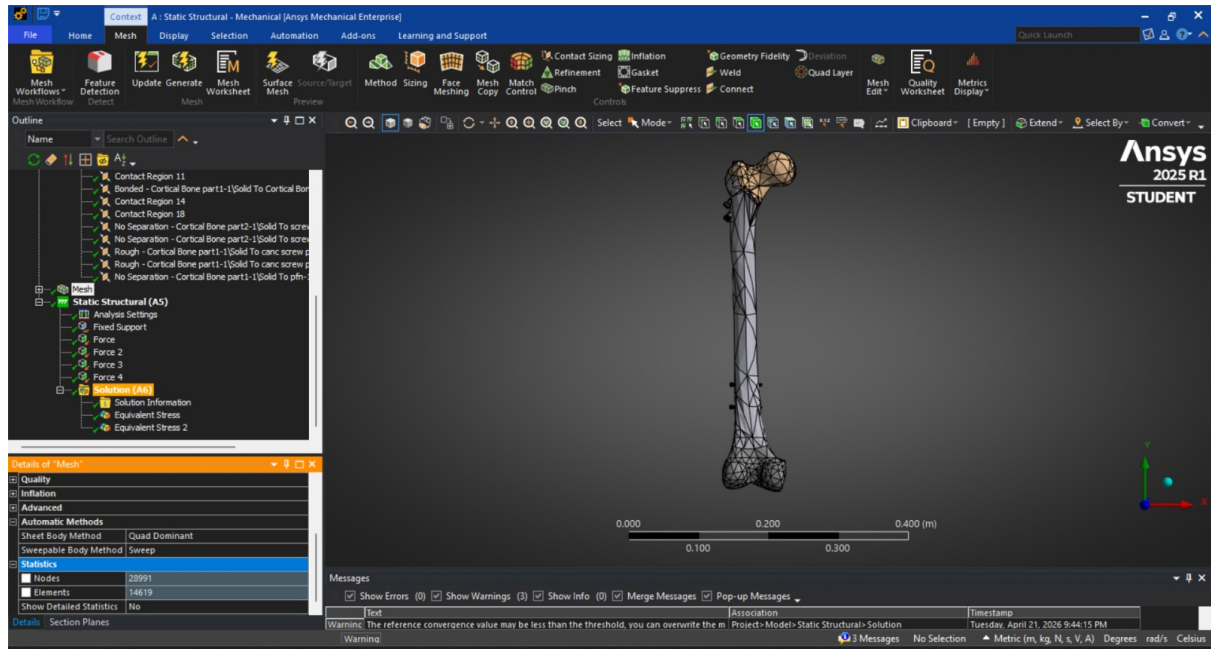


Figure 6.4: Finite Element Mesh of PFNA-II Implant Model

ANNEXURE 5

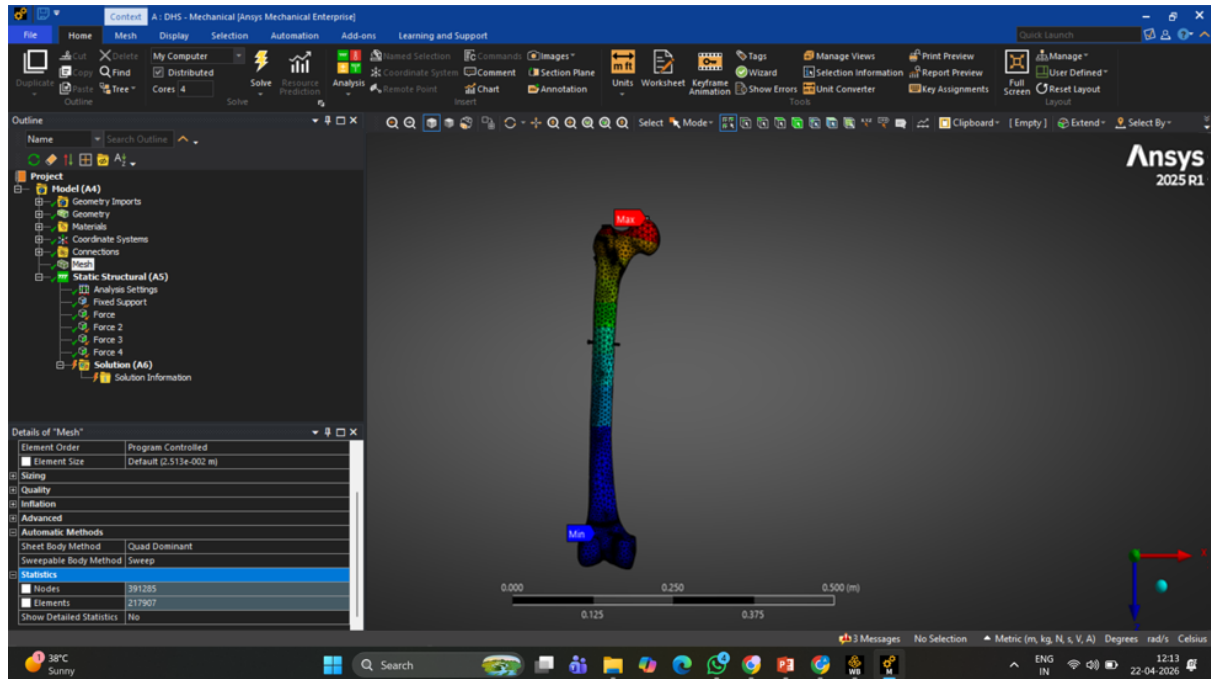


Figure 6.5: Finite Element Mesh of InterTAN Implant Model